

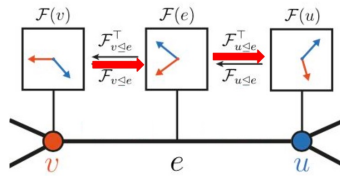


Introduction & Background

- Many physical systems are graphs embedded in the Euclidean space.
- Existing Equivariant GNN** preserve $E(n)$ symmetry but are often **isotropic**, struggling with shear, torsion, vorticity and directional interactions.
- Standard **Sheaf Neural Networks** model anisotropic transport but do not preserve $E(n)$ equivariance.



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How to move a vector x from v to u ?

Moving $x \in \mathcal{F}(v) \rightarrow \mathcal{F}(u)$ is done via maps composition!

$$\mathcal{F}_{u \leftarrow e}^\top \mathcal{F}_{v \leftarrow e} x \in \mathcal{F}(u)$$

What is a Sheaf Neural Network?

- A cellular sheaf equips a graph with local feature spaces $\mathcal{F}(v), \mathcal{F}(e)$ (**stalks**) and edge-dependent **transport maps**.
- The **disagreement** between neighbouring nodes is measured after transporting their representations into a common space.

$$(\delta_{\mathcal{F}} h)_{j \rightarrow i} = \mathcal{F}_{i \leftarrow e} h_i - \mathcal{F}_{j \leftarrow e} h_j$$

- The **Sheaf Laplacian** is:

$$L_{\mathcal{F}}(x)_u = \sum_{u \leftarrow v \in e} \mathcal{F}_{u \leftarrow e}^\top (\mathcal{F}_{u \leftarrow e} x_u - \mathcal{F}_{v \leftarrow e} x_v)$$

Equivariant Sheaf Neural Networks

- Each node stalk contains **invariant scalar features** and $O(n)$ -**covariant vector features**.
- ESNN factorises every edge map into **covariant spatial transport** and **invariant channel mixing**.
- Restriction maps** are constrained to **preserve rotations, reflections, and translations**.

$$\tau_{i \leftarrow j}(V_j) = \sum_{k=1}^K S_{ij}^{(k)} V_j M_{ij}^{(k)}$$

Left multiplication (acts on spatial rows)

$$S_{ij}^{(k)} \in \mathbb{R}^{n \times n}$$

Acts on ambient space

vector channels $\mathbb{R}^{c_v}$ (columns)

$$V_j \in \mathbb{R}^{n \times c_v}$$

ambient space $\mathbb{R}^n$ (rows)

Right multiplication (acts on channel columns)

$$M_{ij}^{(k)} \in \mathbb{R}^{c_v \times c_v}$$

Mixes vector channels

$$V_j \in \mathbb{R}^{n \times c_v}$$

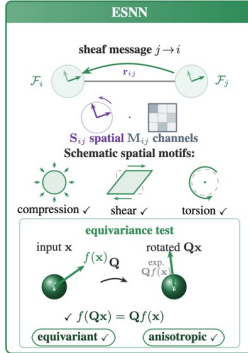
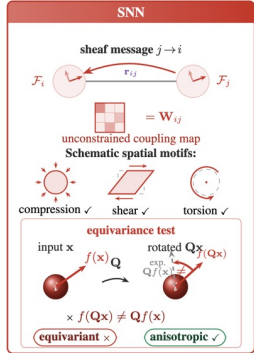
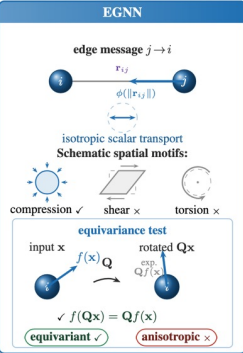
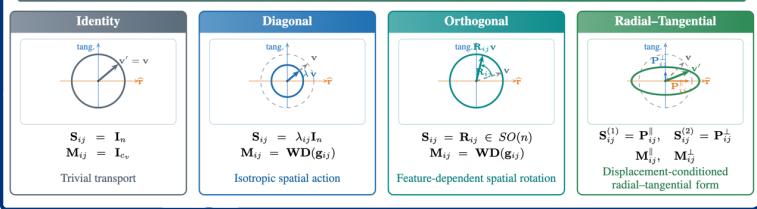
Research Question

Can we combine anisotropic transport of SheafNNs with $E(n)$ -equivariance of geometric GNNs?

Transport Types

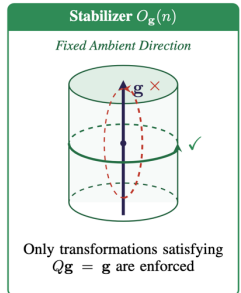
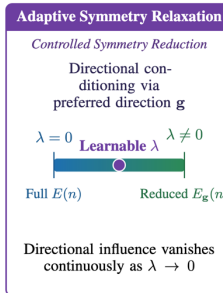
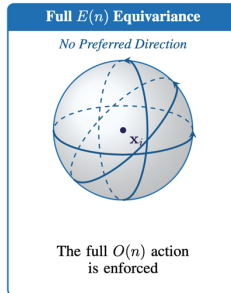
- Increasingly flexible constraints recover existing equivariant models and culminate in maximally expressive first-order anisotropic transport.

Principal ESNN spatial transport families



- Equivariant GNNs** preserve $E(n)$ -symmetry but usually rely on **isotropic radial interactions**.
- Standard Sheaf Neural Networks** support **anisotropic transport**, but unconstrained restriction maps may violate equivariance.
- ESNN combines both properties by constraining spatial transport to be $O(n)$ -covariant.
- ESNN messages can model compression, shear, torsion, and direction-dependent interactions while remaining equivariant.

- Full $E(n)$ -equivariance is appropriate for isolated systems with no preferred direction.
- External fields, such as **gravity** or **uniform flow**, introduce a distinguished spatial axis.
- ESNN learns a relaxation parameter λ , preserving full equivariance when $\lambda \approx 0$.
- When the data require it, the model relaxes to cylindrical $O_g(n)$ -subequivariance, preserving only transformations that leave the field direction g fixed.



Experimental Results

(a) N-body

Method	MSE ↓
Linear	0.0819
SE(3) Transformer	0.0244
Tensor Field Network	0.0155
Graph Neural Network	0.0107
Radial Field	0.0104
EGNN	0.0071
ESNN-Id	0.0060
ESNN-Diag	0.0054
ESNN-Ortho	0.0051
ESNN-RadTan	0.0052

(b) N-body + Gravity - Soft Symmetry

Method	Transport	MSE ↓	max _{i,j} A _{ij} g _{ij}	A _{ij} ↑
Linear	None	0.107100 ± 0.034566	-	-
SE(3) Transformer	ESNN-Ortho	0.125181 ± 0.048970	-	-
Tensor Field Network	ESNN-RadTan	0.101860 ± 0.032444	-	-
Graph Neural Network	ESNN-Diag	0.026545 ± 0.001611	0.287400 ± 0.070406	1.000
Radial Field	ESNN-Ortho	0.019887 ± 0.001452	0.400903 ± 0.072151	1.000
EGNN	ESNN-RadTan	0.021276 ± 0.003345	2.052211 ± 0.633743	1.000
ESNN-Id	ESNN-Diag	0.020838 ± 0.002625	0.346428 ± 0.037274	1.000
ESNN-Diag	ESNN-Ortho	0.019825 ± 0.001611	0.609556 ± 0.177071	1.000
ESNN-Ortho	ESNN-RadTan	0.023381 ± 0.005414	1.379156 ± 0.301397	1.000

MeshGraphNets – Mesh-based Physical Dynamics

Method	CylinderFlow			DeformingPlate			Airfoil		
	1-Step	50-Step	Full	1-Step	50-Step	Full	1-Step	50-Step	Full
MeshGraphNets	2.34 ± 0.12	6.3 ± 0.7	40.88 ± 7.2	0.25 ± 0.05	1.8 ± 0.5	15.1 ± 4.0	314 ± 36	582 ± 37	11529 ± 1203
ESNN-Id	3.96	15.8	69.6	0.15	1.3	7.7	3529	9910	17974
ESNN-Diag	2.82	10.3	37.04	0.17	1.4	9.2	5130	9616	11598
ESNN-Ortho	2.40	7.7	35.94	0.13	1.1	8.1	3014	4722	10328
ESNN-RadTan	3.08	10.9	49.27	0.08	1.0	5.8	2584	3346	7787

ModelNet40 – Classification under Rotations

Method	z/z	SO(3)	SO(3)/SO(3)	Δ _{OOD} ↓
PointNet	85.9	19.6	74.7	66.3
RS-CNN	90.3	48.7	82.6	41.6
DGCNN	90.3	33.8	88.6	56.5
RI-Conv	86.5	86.4	86.4	0.1
GC-Conv	89.0	89.1	89.2	0.1
Luo et al. DGCNN	88.4	88.4	88.4	0.0
LGR-Net	90.9	90.9	91.1	0.0
Li et al. (w/ TTA)	91.6	91.6	91.6	0.0
CRN	91.8	91.8	91.8	0.0
TFN	88.5	85.3	87.6	3.2
VN-PointNet	77.5	77.5	77.2	0.0
VN-DGCNN	89.5	89.5	89.2	0.0
ESNN-Id	84.684	85.737	85.575	1
ESNN-Diag	84.603	84.319	84.400	0.2
ESNN-Ortho	84.927	85.170	84.684	0.2
ESNN-RadTan	85.373	84.643	86.264	0.7

QM9 – Molecular Property Prediction Benchmark

Method	μ	Δμ	χ ₀₀₀₀	χ ₀₀₀₀	μ	C ₆	G	H	H ⁽²⁾	U	U ₀	2PVE
Units	meV	meV	meV	meV	meV	meV	meV	meV	meV	meV	meV	meV
Invariant models												
Commuter	85	61	34	38	38	26	20	21	961	21	22	2.01
NMPP	92	69	43	38	30	40	19	17	180	20	150	
DimeNet++	44	32.6	24.6	19.5	29.7	23	7.56	6.55	331	6.28	6.32	1.21
CoordNet	45	32.4	23.1	19.8	24.5	22	7.08	6.86	259	6.42	6.60	1.20
Spherenet	46	31.1	22.8	18.9	24.5	22	7.78	6.33	268	6.36	6.26	1.12
Scalation-based models												
ChelNet	10	53	33	25	40	27	9	9	610	9	8	1.23
EGNN	71	48	29	25	29	31	12	12	106	12	11	1.59
PathNet	74	45.7	27.6	20.4	12.0	24	13	13	106	12	11	1.59
LEFTNet	48	40	24	18	12	23	7	6	109	7	6	1.33
EQGAT	53	32	20	16	11	24	13	13	106	12	11	1.59
ET	59	36.1	20.3	17.5	11	26	7.62	6.16	33	6.38	6.15	1.84
Geometric	40	33.4	18.4	15.4	10	22	6.13	4.39	38	4.41	4.43	1.28
SciVecNet	39	24.8	18.4	16.3	9.3	23	6.64	5.43	38	5.48	5.43	1.18
Equivariant Sheaf Neural Networks												
ESNN-Id	63.08	65.55	43.13	28.52	58.81	37.83	15.76	10.71	749.44	-	10.54	1.54
ESNN-Diag	62.16	63.69	37.44	21.45	37.35	34.87	11.48	8.271	134.99	11.73	13.57	1.62
ESNN-Ortho	78.42	43.55	28.88	23.17	21.77	36.50	17.76	20.07	323.52	18.59	23.25	1.94
ESNN-RadTan	66.32	39.03	25.16	19.17	16.48	34.87	14.77	13.36	119.11	18.60	18.34	1.48

